Notes on Model Evaluation Metrics:

— XGBoost (eXtreme Gradient Boosting):

- is an advanced implementation of gradient boosting algorithm.
- It's a powerful machine learning algorithm especially popular for structured or tabular data.

— It works by sequentially adding simple models to correct the errors made by previous models.

— It is optimized to be highly computationally efficient.

— It can handle large datasets.

— XGBoost has an in-built routine to handle missing data.

— XGBoost is used for both Regression and Classification.

— How XGBoost Works?

- **Start with a base learner**
- **Calculate the errors**
- **Train the next tree**
- **Repeat the process**
- **Combine the predictions**

Regression:

I. Given data and initial predictions

- start with a default set of predictions (by default set to 0.5 in both classification and regression)

II. Calculating Residuals

III. Building XGBoost Trees

1. **Start with all residuals in the same leaf.**

2. **Calculate the similarity score using the following formula:**

$$\text{Similarity Score} = \frac{(\sum residuals)^2}{len(residuals) + \lambda}$$

Regularization parameter

3. **Finding the best splitting feature and its values**

— If the dataset is huge we can see that this step will end up taking a lot of time. To tackle that we can split using weighted quantiles.

4. **Calculating Gain**
5. **Creating the tree**
6. **Pruning the tree**

— setting gamma to 0 doesnt turn off pruning because there may be nodes with negative gain values, and gain-r will be < 0 in that case, leading to pruning.

7. **Calculating the output Value:**

$$\text{Output Value} = \frac{\sum residuals}{len(residuals) + \lambda}$$

Regularization parameter
(default 0)

IV. Updating the residuals and getting the output

Classification:

I. Given data and initial predictions

II. Calculating Residuals

III. Building XGBoost Trees

1. **Start with all residuals in the same leaf.**
2. **Calculate similarity score:**

$$\text{Similarity Score} = \frac{(\sum residuals)^2}{\sum P_{i-1} * (1 - P_{i-1}) + \lambda}$$

3. **Finding the best splitting feature and its values**
4. **Calculating Gain**
5. **Creating the tree**
6. **Pruning the tree**
7. **Calculating Output:**

$$\text{Output Value} = \frac{\sum(residuals)}{\sum P_{i-1} * (1 - P_{i-1}) + \lambda}$$

IV. Updating the residuals and getting the output

Advantages of XGBoost

- Scalable for large datasets with millions of records.
- Includes feature importance analysis.
- Offers customizable parameters and regularization for fine-tuning.
- Supports parallel processing and GPU acceleration.

Disadvantages of XGBoost

- Sensitive to noise and outliers; careful preprocessing required.
- Can overfit, especially on small datasets or with too many trees.
- Computationally intensive; may not be suitable for resource-limited systems.
- Limited interpretability compared to simpler models.